# Bagging, Multiclass SAMME Boosting, and Bias–Variance on Olivetti Faces

**Rahul Saxena** · Roll No. **M26CSE006**
GitHub repository: **ADD_PUBLIC_REPOSITORY_URL_HERE**
Final commit hash: **ADD_FINAL_COMMIT_HASH_HERE**

**Abstract—** This report studies decision-tree classification on the Olivetti Faces dataset, comparing a single validation-selected tree, bootstrap aggregation, and multiclass SAMME boosting. A fixed identity-stratified split is used, and test-time robustness is assessed by masking a centered 16×16 image region. A repeated-resampling experiment estimates prediction bias and variance across tree depths. **All quantitative result fields remain pending execution and verification**; this draft intentionally does not invent measured values.

## 1. Introduction

Small image datasets can expose the instability of high-capacity decision trees. Bagging reduces sensitivity to the sampled training set by aggregating bootstrap-trained trees, while boosting sequentially emphasizes examples that earlier weak learners misclassify. This assignment evaluates both ensemble strategies and examines the bias–variance trade-off under repeated resampling.

## 2. Dataset and Protocol

We use the 400 grayscale images in the scikit-learn Olivetti Faces dataset. Each image is 64×64 pixels and belongs to one of 40 identities. With seed 26006, six images per identity are assigned to training, two to validation, and two to testing (240/80/80 total). The split indices are saved in *data/olivetti_split_seed_26006.npz*. No PCA or other dimensionality reduction is applied.

For masked testing, pixels in rows 24–39 and columns 24–39 are set to zero in each test image. Hyperparameters are selected using validation accuracy; the test set is reserved for final evaluation. All comparisons should use the same fixed split and configuration.

## 3. Methods

**Decision tree.** Candidate max depths are 2, 4, 8, and unlimited, crossed with minimum leaf sizes 1, 2, and 4. The selected setting is the validation-accuracy winner.

**Bagging.** Each estimator is trained on a bootstrap sample of the original training-set size. The ensemble sizes are $B \in \{1, 5, 10, 25, 50, 100, 200\}$. Report train, validation, clean-test, and masked-test performance where required, as well as distinct bootstrap examples, out-of-bag performance, pairwise disagreement, and vote margin.

**SAMME.** Multiclass AdaBoost is implemented from scratch. At each round, a weak decision tree is fitted using sample weights; its weighted error determines its learner weight, and the sample weights are updated. Depth and number of rounds are selected on the clean validation set.

**Bias–variance.** Repeat training on 30 resampled training sets and compare prediction variability for the prescribed depth settings. The final analysis should state the precise decomposition/estimator used by the code and discuss its limitations on a finite multiclass test set.

## 4. Experimental Results

**Execution status: NOT YET VERIFIED.** Run the reproduction commands and replace every item below with measured outputs from the saved CSV/log files.

- Best tree depth / min leaf: [FILL FROM VALIDATION TABLE]
- Best bagging size and test accuracy: [FILL FROM RESULTS]
- SAMME depth / rounds and test accuracy: [FILL FROM RESULTS]
- Clean vs. masked test accuracy: [FILL FROM RESULTS]
- Bias–variance comparison across depths: [FILL FROM RESULTS]

## 5. Discussion

Use the measured results to answer three questions. First, does bagging improve test performance over the selected single tree, and does its benefit saturate as B increases? Second, does SAMME improve accuracy relative to bagging, and do difficult examples receive increasing weight during training? Third, how does tree depth affect the empirical bias–variance trade-off? For masked images, discuss whether the central occlusion changes predictions systematically and avoid drawing conclusions beyond this dataset and split.

The final discussion must be evidence-led: cite actual values, refer to generated plots/tables, and distinguish observed results from plausible explanations. Do not state that an ensemble is better unless the recorded results support that conclusion.

## 6. Reproducibility

The repository includes source code for Q1–Q3, the latest notebook, JSON configuration, a fixed split file, environment specifications, plotting utilities, and reproduction instructions. The dataset is retrieved through scikit-learn. Run commands from the repository root and retain the generated logs, tables, and figures in the results directory.

Before submission, replace the repository URL and commit hash above, execute all experiments, inspect outputs, add final logs and figures, and regenerate this PDF with the actual results. The final report must be no more than eight pages excluding references; this draft is shorter but is not yet a final results report.

## Generative AI disclosure

ChatGPT assisted with repository organization, documentation, and debugging. The student must verify the code, rerun all experiments, interpret results independently, and follow the complete course AI policy. The policy text provided to the assistant was truncated, so this disclosure must be checked and amended before submission.

## References

[1] AT&T; Laboratories Cambridge, “The Olivetti Faces Dataset,” distributed via scikit-learn’s *fetch_olivetti_faces* interface.
[2] Y. Freund and R. E. Schapire, “A Decision-Theoretic Generalization of On-Line Learning and an Application to Boosting,” 1997.
[3] L. Breiman, “Bagging Predictors,” Machine Learning, 1996.